Trigonometric Functions and Circular Measure

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises**

In a circle of radius 10 cm, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

(a) Use the cosine rule in triangle OAB with $OA = OB = 10$ and $\angle AOB = \frac{2\pi}{3} = 120^\circ$: $AB^2 = 10^2 + 10^2 - 2(10)(10)\cos\left(\frac{2\pi}{3}\right) = 100 + 100 - 200\left(-\frac{1}{2}\right) = 200 + 100 = 300$ Thus $AB = \sqrt{300} = 10\sqrt{3}$ cm. (b) Area of sector OAB: $A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(10^2)\left(\frac{2\pi}{3}\right) = \frac{100\pi}{3}$ cm^2 Area of triangle OAB.

2 **[Foundational]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises**

A circle has radius 8 cm. A sector of this circle has a total perimeter of 28 cm. Find the angle of the sector in radians, and calculate the exact area of the sector.

The perimeter of a sector is given by $P = s + 2r = r\theta + 2r$. Substitute $P = 28$ and $r = 8$: $28 = 8\theta + 2(8) \Rightarrow 28 = 8\theta + 16 \Rightarrow 8\theta = 12 \Rightarrow \theta = \frac{12}{8} = 1.5$ radians The area of the sector is: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(8^2)(1.5) = \frac{1}{2}(64)(1.5) = 32 \times 1.5 = 48$ cm^2

3 **[Foundational]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises**

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

Substitute $\cos(2\theta) = 1 - 2\sin^2\theta$ to express the entire equation in terms of $\sin\theta$: $(1 - 2\sin^2\theta) + 3\sin\theta = 2.$ $-2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0.$ Factor the quadratic in $\sin\theta$: $(2\sin\theta - 1)(\sin\theta - 1) = 0.$ Case 1: $2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$. On $[0, 2\pi)$, $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Case 2: $\sin\theta - 1 = 0 \Rightarrow \sin\theta = 1$. On.

4 **[Foundational]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises**

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

Step 1: Factor the quadratic in $\sin x$: $(2\sin x - 1)(\sin x - 1) = 0.$ Step 2: Solve each factor branch:Branch 1: $2\sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}$. On $[0, 2\pi)$, $x = \frac{\pi}{6}$ (Quadrant I) or $x = \frac{5\pi}{6}$ (Quadrant II).Branch 2: $\sin x - 1 = 0 \Rightarrow \sin x = 1$. On $[0, 2\pi)$, $x = \frac{\pi}{2}$.The complete solution set is $x \in \left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}\right\}$.

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5 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

Expand the double-angle expressions on the left-hand side: LHS $= \frac{2\sin x\cos x}{\sin x} - \frac{2\cos^2 x - 1}{\cos x}$. Cancel $\sin x$ in the first term and combine over a common denominator of $\cos x$:
$= 2\cos x - \frac{2\cos^2 x - 1}{\cos x} = \frac{2\cos^2 x - (2\cos^2 x - 1)}{\cos x}$. $= \frac{2\cos^2 x - 2\cos^2 x + 1}{\cos x} = \frac{1}{\cos x} = \sec x = \text{RHS}.$ ✓

6 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

In ΔABC, $a = 8$, $b = 11$, and $A = 35^\circ$. Determine the number of possible triangles and find the measure of angle B for all valid cases to the nearest tenth of a degree.

Calculate the altitude $h = b\sin A = 11\sin 35^\circ \approx 11(0.5736) = 6.31$. Since $h < a < b$, there are two possible triangles. By the Law of Sines: \sin B = \frac{b\sin A}{a} = \frac{6.31}{8} \approx 0.7887. Second solution (obtuse): $B_2 = 180^\circ - 52.1^\circ = 127.9^\circ$. Check validity: $A + B_2 = 35^\circ + 127.9^\circ = 162.9^\circ 0$. Both $B_1 \approx 52.1^\circ$ and $B_2 \approx 127.9^\circ$ are valid.

7 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Step 1: Determine the rotation angle θ: Here $A = 5$, $B = -4$, and $C = 8$.
$\cot(2\theta) = \frac{A - C}{B} = \frac{5 - 8}{-4} = \frac{-3}{-4} = \frac{3}{4} > 0$. Since $\cot(2\theta) = 3/4$, by right triangle trigonometry $\cos(2\theta) = \frac{3}{5}$. Using half-angle formulas for $\theta \in (0, \pi/4)$: $\cos\theta = \sqrt{\frac{1 + \cos(2\theta)}{2}} = \sqrt{\frac{1 + 3/5}{2}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}},$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Step 1: Express -16 in polar modulus-argument form: $z^4 = -16 = 16(\cos\pi + i\sin\pi)$. Here $r = 16$, $\theta = \pi$, and $n = 4$. Step 2: Apply n-th roots formula: $w_k = \sqrt[4]{16}\left[\cos\left(\frac{\pi + 2\pi k}{4}\right) + i\sin\left(\frac{\pi + 2\pi k}{4}\right)\right] = 2\left[\cos\left(\frac{(2k+1)\pi}{4}\right) + i\sin\left(\frac{(2k+1)\pi}{4}\right)\right],\ k \in \{0, 1, 2, 3\}.$ Step 3: Evaluate for each k:$k = 0$.

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9 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

Equate the two equations: $2\sin\theta = 2\cos\theta \Rightarrow \tan\theta = 1$. On $[0, 2\pi)$, $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.For $\theta = \frac{\pi}{4}$: $r = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$, giving point $\left(\sqrt{2}, \frac{\pi}{4}\right)$.For $\theta = \frac{5\pi}{4}$: $r = 2\sin\left(\frac{5\pi}{4}\right) = -\sqrt{2}$. However, $\left(-\sqrt{2}, \frac{5\pi}{4}\right)$ represents the identical geometric point as $\left(\sqrt{2}, \frac{\pi}{4}\right)$.Next, check the pole.

10 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.

1. Vertical height descended: $h = 8\sin 30^\circ = 8(0.5) = 4\ \text{m}$. Initial gravitational potential energy: $\text{PE} = mgh = 5(9.8)(4) = 196\ \text{J}$ 2. Normal reaction: $R = mg\cos 30^\circ = 5(9.8)\left(\frac{\sqrt{3}}{2}\right) = 24.5\sqrt{3}\ \text{N}$. Friction force: $F = \mu R = 0.2(24.5\sqrt{3}) = 4.9\sqrt{3}\ \text{N}$. Work done against friction over distance $d = 8\ \text{m}$: $W_{\text{fric}} = F \times d = (4.9\sqrt{3}) \times 8 = 39.2\sqrt{3} \approx 67.896\ \text{J}$ 3. By.

11 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Determine whether the lines $\mathbf{r}_1 = (1; 2; -1) + \lambda(2; 1; 3)$ and $\mathbf{r}_2 = (0; 5; 4) + \mu(-1; 2; 1)$ intersect or are skew.

1. Check parallelism: Direction vectors are $\mathbf{d}_1 = (2; 1; 3)$ and $\mathbf{d}_2 = (-1; 2; 1)$. Since $\frac{2}{-1} eq \frac{1}{2}$, they are not scalar multiples, so the lines are not parallel. 2. Equate components: $1 + 2\lambda = -\mu \Rightarrow 2\lambda + \mu = -1 \quad (1)$ $2 + \lambda = 5 + 2\mu \Rightarrow \lambda - 2\mu = 3 \quad (2)$ $-1 + 3\lambda = 4 + \mu \Rightarrow 3\lambda - \mu = 5 \quad (3)$ Add (1) and (3): $5\lambda = 4 \Rightarrow \lambda = \frac{4}{5}$. From.

12 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

Step 1: Compute dot product and squared magnitude: $\mathbf{u} \cdot \mathbf{v} = (6)(3) + (2)(4) = 18 + 8 = 26$. $||\mathbf{v}||^2 = 3^2 + 4^2 = 9 + 16 = 25$. Step 2: Vector projection (parallel component): $\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{26}{25}\langle 3, 4 \rangle = \left\langle \frac{78}{25}, \frac{104}{25} \right\rangle = \langle 3.12, 4.16 \rangle$. Step 3: Orthogonal component.

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13 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A stone is thrown vertically upwards with speed 15 m/s from the top of a cliff 40 m above sea level. Taking $g = 9.8$ m/s^2 and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

(a) At maximum height, $v = 0$. $u = +15$ m/s, $a = -9.8$ m/s^2: $v^2 = u^2 + 2as \Rightarrow 0 = 15^2 + 2(-9.8)s \Rightarrow 19.6s = 225 \Rightarrow s = \frac{225}{19.6} \approx 11.48$ m ≈ 11.5 m (b) At the sea, the displacement relative to the launch point is $s = -40$ m. Use $s = ut + \frac{1}{2}at^2$: $-40 = 15t + \frac{1}{2}(-9.8)t^2 \Rightarrow -40 = 15t - 4.9t^2 \Rightarrow 4.9t^2 - 15t - 40 = 0$ Solve via quadratic formula.

14 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

Step 1: Form vectors $\vec{AB}$ and $\vec{AC}$: $\vec{AB} = \langle 2-1, 4-2, 1-3 \rangle = \langle 1, 2, -2 \rangle$. $\vec{AC} = \langle -1-1, 0-2, 5-3 \rangle = \langle -2, -2, 2 \rangle$. Step 2: Compute normal vector via cross product: $\mathbf{n} = \vec{AB} \times \vec{AC} = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, 2, -2; -2, -2, 2| = \mathbf{i}(4-4) - \mathbf{j}(2-4) + \mathbf{k}(-2-(-4)) = 0\mathbf{i} + 2\mathbf{j} + 2\mathbf{k} = \langle 0, 2, 2 \rangle$. Simplifying by scalar.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Find the binomial expansion of $\frac{1}{\sqrt{4-x}}$ in ascending powers of x up to and including the term in x^2. State the set of values of x for which the expansion is valid.

Rewrite with negative fractional index and factor out 4: $(4-x)^{-1/2} = 4^{-1/2}(1 - \frac{x}{4})^{-1/2} = \frac{1}{2}(1 - \frac{x}{4})^{-1/2}$ Apply $(1+y)^n = 1 + ny + \frac{n(n-1)}{2}y^2 + \cdots$ with $n = -\frac{1}{2}$ and $y = -\frac{x}{4}$: $(1 - \frac{x}{4})^{-1/2} = 1 + (-\frac{1}{2})(-\frac{x}{4}) + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2}(-\frac{x}{4})^2 + \cdots$.

16 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Use the substitution $u = 2x + 1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx$.

Let $u = 2x + 1$, so $du = 2\,dx \Rightarrow dx = \frac{1}{2}\,du$, and $x = \frac{u-1}{2}$. Change limits: $x = 0 \Rightarrow u = 1, \quad x = 4 \Rightarrow u = 9$ Transform the integral: $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx = \int_1^9 \frac{\frac{u-1}{2}}{\sqrt{u}} \cdot \frac{1}{2}\,du = \frac{1}{4}\int_1^9 (u^{1/2} - u^{-1/2})\,du$ $= \frac{1}{4}\left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^9 = \frac{1}{4}\left(\left(\frac{2}{3}(27) - 2(3)\right) - \left(\frac{2}{3}(1) - 2(1)\right)\right)$.

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17 **[Challenging]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises** • **Cross-Lesson: 8.01 Integration and Kinematics**

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

(a) Under H_0, $X \sim \text{Po}(6)$. Calculate cumulative probabilities: $P(X = 0) = e^{-6} \approx 0.00248$ $P(X \le 1) = e^{-6}(1 + 6) = 7e^{-6} \approx 0.01735$ $P(X \le 2) = e^{-6}(1 + 6 + 18) = 25e^{-6} \approx 0.06197$ Since $P(X \le 1)0.05$, the critical region is $X \le 1$ (i.e. $X = 0$ or 1). (b) Actual significance level = $P(X \le 1 \mid \lambda = 6) = 7e^{-6} \approx 0.0174$ (or 1.74%). (c) Type II error occurs when we fail to reject H_0 when $\lambda = 3$. We fail to reject when.

18 **[Challenging]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises** • **Cross-Lesson: 1.01 Functions, Modulus and Quadratics**

A small particle of weight 24 N is held in equilibrium by two light inextensible strings. String A makes an angle of 30° with the upward vertical, and string B makes an angle of 60° with the upward vertical. Find the tensions T_A and T_B in the two strings.

Notice that the angle between string A and string B is $30^\circ + 60^\circ = 90^\circ$, so the strings are perpendicular. Resolve horizontally: $T_A \sin 30^\circ = T_B \sin 60^\circ \Rightarrow T_A(0.5) = T_B\left(\frac{\sqrt{3}}{2}\right) \Rightarrow T_A = \sqrt{3}T_B$ Resolve vertically: $T_A \cos 30^\circ + T_B \cos 60^\circ = 24 \Rightarrow T_A\left(\frac{\sqrt{3}}{2}\right) + T_B(0.5) = 24$ Substitute $T_A = \sqrt{3}T_B$.

19 **[Challenging]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises** • **Cross-Lesson: 2.01 Simultaneous Equations and Inequalities**

Prove by mathematical induction that for all integers $n \ge 1$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.

1. Base Step ($n = 1$): LHS $= 2(1) - 1 = 1$, RHS $= 1^2 = 1$. Since LHS = RHS, the base case $P(1)$ is true. 2. Inductive Hypothesis: Assume $P(k)$ is true for some integer $k \ge 1$: $1 + 3 + 5 + \ldots + (2k - 1) = k^2$. 3. Inductive Step ($n = k + 1$): Show that $1 + \ldots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2$: LHS $= [1 + 3 + \ldots + (2k - 1)] + (2k + 1)$. By the inductive hypothesis, substitute k^2.

20 **[Challenging]** • **Target: Trigonometric Functions & Circular Measure Practice Exercises** • **Cross-Lesson: 3.01 Indices, Surds and Polynomials**

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2 - 4 = (x - 2)(x + 2)$. The numerator $3x^2 + 1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{3 + 1/x^2}{1 - 4/x^2} = \frac{3}{1} = 3$. The horizontal asymptote is $y = 3$. Notice tha.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A circle sector has radius \(r = 6 \text{ cm} \) and central angle \(\theta = \frac{\pi}{3} \text{ rad} \). What is the arc length of the sector?

a. **\(\frac{6\pi}{3} \text{ cm} \) ✓**

b. \(\frac{36\pi}{6} \text{ cm} \)

c. \(6\pi \text{ cm} \)

d. \(\frac{6\pi}{6} \text{ cm} \)

Arc length formula in radians: \(s = r\theta = (6)\left(\frac{\pi}{3}\right) = \frac{6\pi}{3} \text{ cm} \).

22 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.

a. **r = 9,** $\theta = \frac{2\pi}{3}$ ✓

b. r = 6, $\theta = \pi$

c. r = 3, $\theta = 2\pi$

d. r = 18, $\theta = \frac{\pi}{3}$

The formula for the arc length of a sector is $L = r\theta$, where L is the arc length, r is the radius, and θ is the angle in radians. We are given that $L = 6\pi$. So, we have the equation: $r\theta = 6\pi$ (1) The formula for the area of a sector is $A = \frac{1}{2}r^2\theta$, where A is the area, r is the radius, and θ is the angle in radians. We are given that $A = 27\pi$. So, we have the equation: $\frac{1}{2}r^2\theta = 27\pi$ (2) We can rewrite equation (2) as $r(r\theta) = 54\pi$. Sub.

23 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is 6 cm and the radius is 3 cm, find the angle θ in radians.

a. $\frac{1}{2}$ radians

b. 2 **radians** ✓

c. $\frac{2}{3}$ radians

d. $\frac{3}{2}$ radians

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is 2 radians.

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24 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector has radius 6 cm and angle 0.3 rad. Find the arc length.

a. **9/5** ✓

b. 63/10

c. 14/5

d. 27/5

Applying Circular Measure principles, $s = r\theta = 6 \times 0.3 = 9/5$ cm Therefore, the verified correct option is 9/5.

25 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector OAB has radius 8 cm and angle $\angle AOB = 1.2$ rad. Find the area of the minor segment cut off by chord AB, leaving your answer in exact form.

a. $32(1.2 - \sin(1.2))$ ✓

b. $64(1.2 - \sin(1.2))$

c. $33(1.2 - \sin(1.2))$

d. $32(1.2 + \sin(1.2))$

Segment $=$ Sector $-$ Triangle $= \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta = 32(1.2 - \sin(1.2))$ cm^2

26 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector has radius 10 cm and angle 1.7 rad. Find its area.

a. 170

b. 95

c. **85** ✓

d. 17

$A = \frac{1}{2}r^2\theta = \frac{1}{2}(10)^2(1.7) = 85$ cm^2

27 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector has radius 5 cm and angle 0.8 rad. Find the perimeter of the sector.

a. **14** ✓

b. 9

c. 4

d. 15

$P = 2r + r\theta = 2(5) + (5)(0.8) = 14$ cm

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28 [Foundational] • Target: Trigonometric Functions & Circular Measure Practice Exercises

A sector of a circle with radius r cm has a perimeter of P cm. The area of the sector is A cm^2. Prove that $P^2 \geq 16A$.

a. **The inequality holds true as shown in the derivation.** ✓

b. The inequality holds true only for specific angles.

c. The inequality fails for angles greater than π radians.

d. The inequality fails for angles less than 2 radians.

Let the angle of the sector be θ radians. The arc length of the sector is $r\theta$. The perimeter $P = 2r + r\theta$. The area $A = \frac{1}{2}r^2\theta$. We need to prove $P^2 \geq 16A$. Substituting the expressions for P and A: $(2r + r\theta)^2 \geq 16\left(\frac{1}{2}r^2\theta\right)$ $r^2(2 + \theta)^2 \geq 8r^2\theta$ Since r^2 is positive, we can divide both sides by r^2: $(2 + \theta)^2 \geq 8\theta$ $4 + 4\theta + \theta^2 \geq 8\theta$ $\theta^2 - 4\theta + 4 \geq 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

If $\sin\theta = -\frac{20}{29}$ and θ is in Quadrant 3, find $\cos\theta$.

a. $\frac{29}{21}$

b. $\frac{21}{29}$

c. $-\frac{21}{29}$ ✓

d. $\frac{20}{29}$

By the Pythagorean identity, $\cos\theta = \pm\sqrt{1 - \sin^2\theta} = \pm\frac{21}{29}$. In Quadrant 3, cosine is negative, so $\cos\theta = -\frac{21}{29}$.

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30 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Find the period and phase shift of $y = \tan(1/2x - 1)$.

a. Period 2π, phase shift 0

b. Period 2π, phase shift 2 units left

c. Period π, phase shift 2 units right

d. **Period 2π, phase shift 2 units right** ✓

For $\tan(Bx - C)$: period $= \frac{\pi}{B} = 2\pi$, phase shift $= \frac{C}{B} = 2$ units right.

31 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Find the reference angle for 220°.

a. 60°

b. **40°** ✓

c. 50°

d. 140°

220° lies in Quadrant 3; the reference angle is 40°.

32 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

What are the maximum and minimum values of $y = 3\sin(x)$?

a. Max 2, Min -2

b. **Max 3, Min -3** ✓

c. Max 0, Min 0

d. Max 4, Min -4

Since $-1 \le \sin(x) \le 1$, y ranges from $-3 = -3$ to $3 = 3$.

33 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

What is the amplitude and period of $y = 5\sin(3x) - 3$?

a. Amplitude -3, period $\frac{2}{3}\pi$

b. **Amplitude 5, period $\frac{2}{3}\pi$** ✓

c. Amplitude 5, period $\frac{3}{2}\pi$

d. Amplitude 5, period 2π

Amplitude $= |A| = 5$. Period $= \frac{2\pi}{3} = \frac{2}{3}\pi$.

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34 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Convert 150° to radians.

a. $\frac{5\pi}{6}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{\pi}{6}$

d. $\frac{7\pi}{4}$

$150^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{6}$.

35 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Convert 270° to radians, giving your answer in terms of π.

a. $\frac{\pi}{2}$

b. π

c. $\frac{3\pi}{2}$ ✓

d. $\frac{\pi}{6}$

$270^\circ = 270 \times \frac{\pi}{180} = \frac{3\pi}{2}$

36 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Convert $\frac{3\pi}{2}$ radians to degrees.

a. 30°

b. 180°

c. 90°

d. 270° ✓

$\frac{3\pi}{2} \times \frac{180}{\pi} = 270^\circ$

37 [Intermediate] • Target: Trigonometric Functions & Circular Measure Practice Exercises

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?

a. $\cos\theta$ ✓

b. $\sin\theta$

c. $\tan\theta$

d. 1

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle θ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point P is $\cos\theta$.

39 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$ ✓

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

We are given $\cos\alpha = \frac{3}{5}$ and α is in the first quadrant. Using the identity $\sin^2\alpha + \cos^2\alpha = 1$, we find $\sin\alpha = \sqrt{1 - (\frac{3}{5})^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ (since α is in the first quadrant, $\sin\alpha$ is positive). We are given $\sin\beta = -\frac{4}{5}$ and β is in the third quadrant. Using the identity $\sin^2\beta + \cos^2\beta = 1$, we find.

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40 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

What are the values of x that satisfy the equation $|3x-2|=2|x+1|$?

a. $x=0$ **and** $x=4$ ✓

b. $x=0$ and $x=-4$

c. $x=\frac{1}{5}$ and $x=4$

d. $x=\frac{1}{4}$ and $x=\frac{3}{2}$

To solve the modulus equation $|3x-2|=2|x+1|$, we can either consider the linear cases or square both sides. **Method 1: Considering linear cases** The equation $|A|=|B|$ (where $B=2(x+1)$) is equivalent to $A=B$ or $A=-B$: - Case 1: $3x-2=2(x+1)\Rightarrow 3x-2=2x+2\Rightarrow x=4$ - Case 2: $3x-2=-2(x+1)\Rightarrow 3x-2=-2x-2\Rightarrow 5x=0\Rightarrow x=0$ **Method 2: Squaring both sides** Since both sides are non-negative, we can square both sides.

41 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Consider the rational function $f(x)=\frac{ax^2-4x+1}{x^2-bx+2}$. The graph of $y=f(x)$ has a vertical asymptote at $x=1$ and a horizontal asymptote at $y=2$. If $f(0)=\frac{1}{2}$, find the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

For a rational function $f(x)=\frac{P(x)}{Q(x)}$, vertical asymptotes occur at the zeros of the denominator $Q(x)$ that are not also zeros of the numerator $P(x)$. Given a vertical asymptote at $x=1$, we know that $x=1$ is a root of the denominator $x^2-bx+2=0$. Therefore, $1^2-b(1)+2=0$, which simplifies to $1-b+2=0$, so $b=3$. The denominator is $x^2-3x+2=(x-1)(x-2)$. The horizontal asymptote is determined by the ratio of the leading coefficients of the numerator and denominator if t.

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42 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

The function f is defined for $x \in \mathbb{R}, x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x \neq 2$?

a. $\frac{3x+1}{x-2}$ ✓

b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

To find the inverse function $f^{-1}(x)$, set $y = f(x)$ and rearrange to express x in terms of y: $y = \frac{2x+1}{x-3}$ Multiply both sides by $x - 3$: $y(x-3) = 2x + 1$ $yx - 3y = 2x + 1$ Collect all terms containing x on one side and constant/independent terms on the other: $yx - 2x = 3y + 1$ Factor out x: $x(y-2) = 3y + 1$ Divide by $y - 2$: $x = \frac{3y+1}{y-2}$ Interchanging the variables gives: $f^{-1}(x) = \frac{3x+1}{x-2}$ Common Misconceptions: - Choice.

43 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. **a = 5, b = 3** ✓

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

For a rational function to have a removable discontinuity at $x = c$, the numerator must have a factor of $(x - c)$ that cancels with a factor in the denominator. Given a removable discontinuity at $x = 2$, the numerator $x^2 - ax + 6$ must have $x = 2$ as a root. Thus, $2^2 - a(2) + 6 = 0$, which simplifies to $4 - 2a + 6 = 0$, so $10 - 2a = 0$, and $a = 5$. The numerator can be factored as $(x - 2)(x - 3)$ since $2 \times 3 = 6$ and $2 + 3 = 5$, so $a = 5$. For the function to have a vertical asymptote at $x = 3$, the.

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44 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

The problem states that point $P(x, y)$ lies on the unit circle, meaning $x^2 + y^2 = 1$. We are given that the y-coordinate of P is $y = \frac{3}{5}$. Substituting this into the unit circle equation, we get $x^2 + (\frac{3}{5})^2 = 1$. This simplifies to $x^2 + \frac{9}{25} = 1$. Subtracting $\frac{9}{25}$ from both sides gives $x^2 = 1 - \frac{9}{25} = \frac{16}{25}$. Taking the square root of both sides, we get $x = \pm\frac{4}{5}$. The problem also states that point P is in the second quadrant, where the x.

45 [Challenging] • Target: Trigonometric Functions & Circular Measure Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers. Which of the following represents the graph of $y = f^{-1}(x)$?

a. Graph A

b. **Graph B** ✓

c. Graph C

d. Graph D

The graph of the inverse function $y = f^{-1}(x)$ is obtained by reflecting the graph of $y = f(x)$ across the line $y = x$. The provided graph of $f(x)$ is a parabola opening upwards with its vertex at $(2, -1)$. Graph A shows a function that is not a reflection of $f(x)$ across $y = x$. Graph B correctly shows the reflection of $y = f(x)$ across the line $y = x$. For example, the point $(0, 3)$ on $f(x)$ corresponds to the point $(3, 0)$ on $f^{-1}(x)$. The vertex $(2, -1)$ on $f(x)$ corresponds to the point.

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